- For a Gaussian beam (whose phase front is plane at $z = 0$), the variation of the spot size is given by

$$w(z) \approx w_0\left[1+\frac{\lambda^2 z^2}{\pi^2 w_0^4}\right]^{1/2}$$

where $w_0$ is the spot size at $z = 0$. For large values of $z$:

$$w(z) \approx \frac{\lambda z}{\pi w_0}$$

which shows that the width increases linearly with $z$. We define the diffraction angle as

$$\tan\theta = \frac{w(z)}{z} \approx \frac{\lambda}{\pi w_0}$$

showing that the rate of increase in width is proportional to the wavelength and inversely proportional to the initial width of the beam; this is characteristic of diffraction. The corresponding radius of curvature of the wavefront is given by

$$R(z) \approx z\left[1+\frac{\pi^2 w_0^4}{\lambda^2 z^2}\right]$$

- For a plane wave incident normally on a straight edge, the intensity variation on a screen (at a distance $d$ from the straight edge) is given by

$$I = \frac{1}{2}I_0\left[\left\{\frac{1}{2}-C(v_0)\right\}^2+\left\{\frac{1}{2}-S(v_0)\right\}^2\right]$$

where $I_0$ is the intensity in the absence of the straight edge,

$$v_0 = -\sqrt{\frac{2}{\lambda d}}\,y$$

$y$ being the distance from the edge of the geometrical shadow and

$$C(x) = \int_0^x \cos\left(\tfrac{1}{2}\pi u^2\right)du$$

and

$$S(x) = \int_0^x \sin\left(\tfrac{1}{2}\pi u^2\right)du$$

are known as Fresnel integrals. The intensity monotonically goes to zero as we go deep inside the geometrical shadow. As we move away from the edge of the geometrical shadow to the illuminated region, one obtains maxima at $v_0 \approx -1.22$ ($I \approx 1.37\ I_0$), $-2.34$ ($I \approx 1.20\ I_0$), $-3.08$ ($I \approx 1.15\ I_0$) … and minima at $v_0 \approx -1.87$ ($I \approx 0.78\ I_0$), $-2.74$ ($I \approx 0.84\ I_0$), $-3.39$ ($I \approx 0.87\ I_0$), ...

- For a plane wave incident normally on a long narrow slit of width $b$, the intensity variation on a screen (at a distance $d$ from the slit) is given by

$$I = \frac{1}{2}\,I_0\,[\{C(v_2+v_1)-C(v_2-v_1)\}^2 + \{S(v_2+v_1)-S(v_2-v_1)\}^2]$$

where

$$v_1 = \sqrt{\frac{2}{\lambda d}}\,\frac{b}{2};\ v_2 = \sqrt{\frac{2}{\lambda d}}\,y$$

and $y$ is the distance from the midpoint of the edges of the geometrical shadow. As $v_1$ becomes large, we obtain the intensity distribution corresponding to two straight edges and for $v_1 \to 0$ we get the Fraunhofer diffraction pattern.

## Problems

**20.1** Consider a plane wave of wavelength $6 \times 10^{-5}$ cm incident normally on a circular aperture of radius 0.01 cm. Calculate the positions of the brightest and the darkest points on the axis.

[**Ans:** $d \approx 1.67$ cm, 0.56 cm, 0.33 cm, … (Maxima); $d \approx 0.83$ cm, 0.42 cm, …(Minima)]

**20.2** What would happen if the circular aperture in Problem 20.1 is replaced by a circular disc of the same radius?

**20.3** **(a)** A plane wave ($\lambda = 6 \times 10^{-5}$ cm) is incident normally on a circular aperture of radius $a$.

**(i)** Assume $a = 1$ mm. Calculate the values of $z$ (on the axis) for which maximum intensity will occur. Plot the intensity as a function of $z$ and interpret physically.

**(ii)** Assume $z = 50$ cm. Calculate the values of $a$ for which minimum intensity will occur on the axial point. Plot the intensity variation as a function of $a$ and interpret physically.

**(b)** Repeat the calculations for $\lambda = 5 \times 10^{-5}$ cm and discuss chromatic aberration of a zone plate.

[**Ans:** (i) (a) $z \approx 166.7$ cm, 55.6 cm, 33.3 cm,… (maxima); (b) Minimum intensity will occur when $a \approx 0.0775$ cm, 0.110 cm, 0.134 cm, …]

**20.4** Consider a circular aperture of diameter 2 mm illuminated by a plane wave. The most intense point on the axis is at a distance of 200 cm from the aperture. Calculate the wavelength.

[**Ans.** $5 \times 10^{-5}$ cm]

**20.5** If a zone-plate has to have a principle focal length of 50 cm corresponding to $\lambda = 6 \times 10^{-5}$ cm, obtain an expression for the radii of different zones. What would be its principle focal length for $\lambda = 5 \times 10^{-5}$ cm?

[$(\sqrt{0.3n})$ mm, 60 cm]

**20.6** In a zone-plate, the second, fourth, sixth…zones are blackened; what would happen if instead the 1st, 3rd, 5th, etc., zones were blackened?

**20.7** **(a)** A plane wave is incident normally on a straight edge (see Fig. 20.21). Show that the field at an arbitrary point $P$ is given by

$$u(P) = \frac{1-i}{2}u_0\left[\left\{\frac{1}{2}-C(v_0)\right\}+i\left\{\frac{1}{2}-S(v_0)\right\}\right]$$

where

$$v_0 = -\sqrt{\frac{2}{\lambda d}}\,y.$$

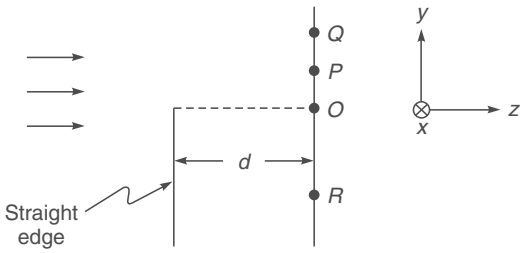



**Fig. 20.21**

**(b)** Assume $\lambda_0 = 5000$ Å and $d = 100$ cm. Using Table 20.1, write approximately the values of $I/I_0$ at the points $O$, $P$ ($y = 0.5$ mm), $Q$ ($y = 1$ mm) and $R$ ($y = -1$ mm) where $O$ is at the edge of the geometrical shadow.

[**Ans:** (b) $I/I_0 \approx 1.26, 0.24, 0.01$]

**20.8** Consider a straight edge being illuminated by a parallel beam of light with $\lambda = 6 \times 10^{-5}$ cm. Calculate the positions of the first two maxima and minima on a screen at a distance of 50 cm from the edge.

[**Ans:** the first two maxima occur at $y \approx 0.0473$ cm and 0.0906 cm. The first two minima occur at $y \approx 0.0724$ cm and 0.1061 cm.]

**20.9** In a straight edge diffraction pattern, one observes that the most intense maximum occurs at a distance of 1 mm from the edge of the geometrical shadow. Calculate the wavelength of light, if the distance between the screen and the straight edge is 300 cm.

[**Ans.** ≈ 4480 Å]

**20.10** In a straight edge diffraction pattern, if the wavelength of the light used is 6000 Å and if the distance between the screen and the straight edge is 100 cm, calculate the distance between the most intense maximum and the next maximum. Find approximately the distance in centimeters inside the geometrical shadow where $I/I_0 = 0.1$.

[**Ans:** $y \approx 0.027$ cm]

**20.11** Consider a plane wave falling normally on a narrow slit of width 0.5 mm. If the wavelength of light is $6 \times 10^{-5}$ cm, calculate the distance between the slit and the screen so that the value of $v_1$ would be 0.5, 1.0, 1.5 and 5.0 (see Figs. 20.16–20.19). Discuss the transition to the Fraunhofer region.

**20.12** Consider the Fresnel diffraction pattern produced by a plane wave incident normally on a slit of width $b$. Assume $\lambda = 5 \times 10^{-5}$ cm, $d = 100$ cm. Using Table 20.1, approximately calculate the intensity values (for $b = 0.1$ cm) at $y = 0, \pm 0.05$ cm, $\pm 0.1$ cm. Repeat the analysis for $b = 5$ cm.

[**Ans:** At $y = 0$, $I/I_0 \approx 1.60$; at $y = \pm 0.05$ cm, $I/I_0 \approx 0.356$; at $y = \pm 0.01$ cm, $I/I_0 \approx 0.01685$].

**20.13** **(a)** The output of a He–Ne laser ($\lambda = 6328$ Å) can be assumed to be Gaussian with plane phase front. For $w_0 = 1$ mm and $w_0 = 0.2$ mm, calculate the beam diameter at $z = 20$ m.

**(b)** Repeat the calculation for $\lambda = 5000$ Å and interpret the results physically.

[**Ans:** (a) 0.83 cm and 4.0 cm]

**20.14** A Gaussian beam is coming out of a laser. Assume $\lambda = 6000$ Å and that at $z = 0$, the beam width is 1 mm and the phase front is plane. After traversing 10 m through vacuum what will be **(a)** the beam width and **(b)** the radius of curvature of the phase front.

[**Ans:** $2w \approx 0.77$ cm; $R(z) \approx 1017$ cm]

**20.15** A plane wave of intensity $I_0$ is incident normally on a circular aperture as shown in Fig. 20.22. What will be the intensity on the axial point $P$? [**Ans:** $3I_0$]

[**Hint:** You may use Eq. (20.27)]

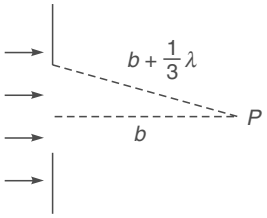



**Fig. 20.22**

**20.16** Show that a phase variation of the type

$$\exp\left[ikz + \frac{ik(x^2 + y^2)}{2R(z)}\right]$$

represents a diverging spherical wave of radius $R$.

**20.17** The output of a semiconductor laser can be approximately described by a Gaussian function with two different widths along the transverse ($w_T$) and lateral ($w_L$) directions as

$$\psi(x, y) = A \exp\left(-\frac{x^2}{w_L^2} - \frac{y^2}{w_T^2}\right)$$

where $x$ and $y$ represent axes parallel and perpendicular to the junction plane. Typically $w_T \approx 0.5$ $\mu$m and $w_L = 2$ $\mu$m. Discuss the far field of this beam (see Fig. 20.23).

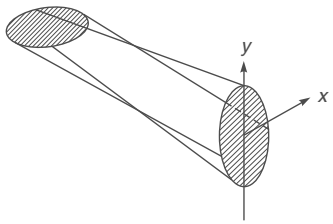



**Fig. 20.23**